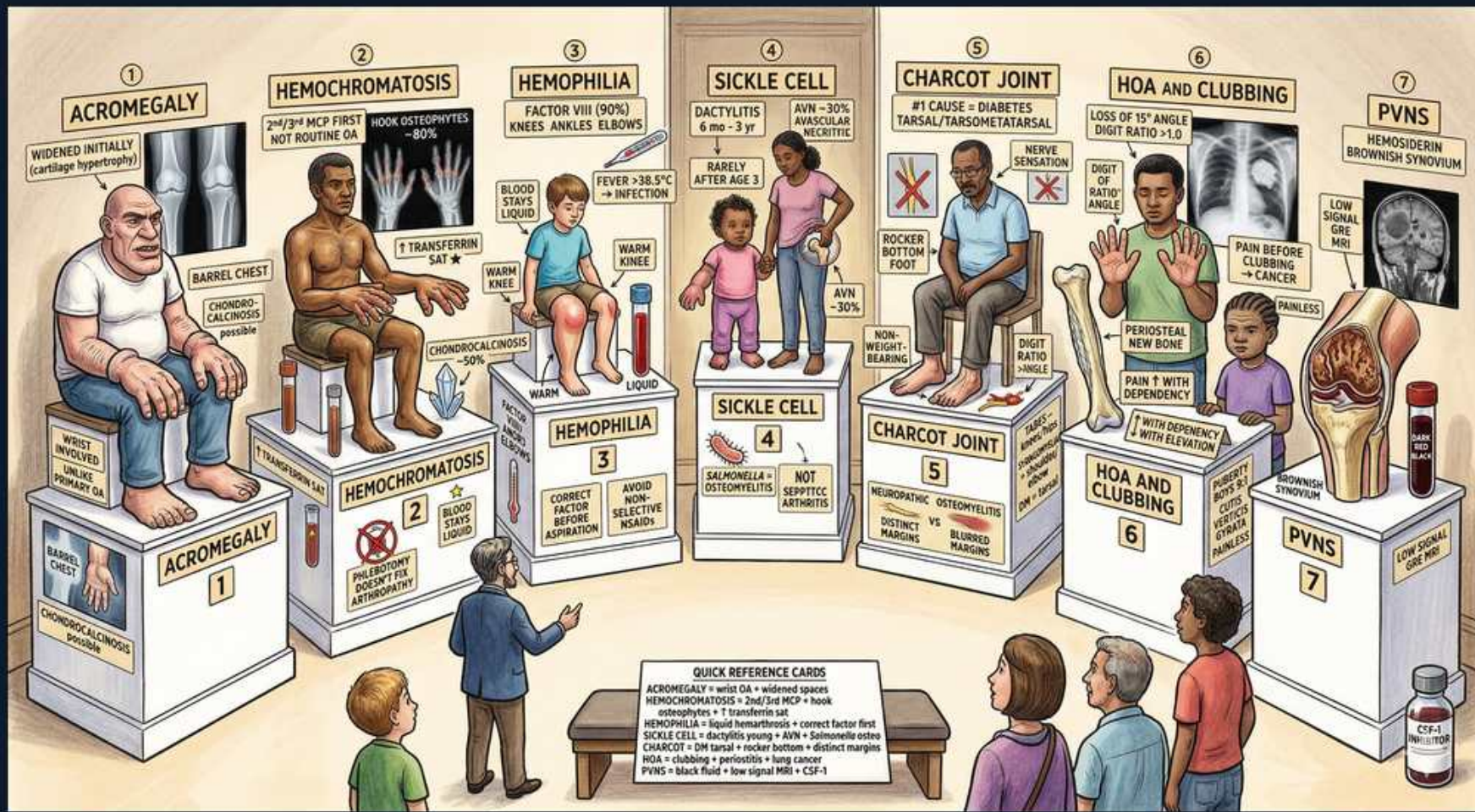


Arthritis — Approach



1. Quick reference card

WHAT YOU SEE

'QUICK REFERENCE CARD' at front

WHAT IT ENCODES

The systematic approach to any arthritis answers four questions before ordering tests: (1) inflammatory vs non-inflammatory, (2) acute (<6 weeks) vs chronic, (3) monoarticular vs oligo/polyarticular, and (4) symmetric vs asymmetric. Inflammatory clues are morning stiffness >60 min, improvement with use, warmth/erythema, and elevated ESR/CRP; non-inflammatory (e.g., OA) has stiffness <30 min and pain worse with use. Synovial fluid analysis then confirms the category. Mnemonic: 'PAMS' — Pattern, Acuity, Mono/poly, Symmetry.

BOARD TRAP

The board wants pattern recognition PLUS synovial fluid, not imaging alone — ordering an MRI of an acute hot monoarthritis instead of aspirating is the wrong first step and delays diagnosis of septic arthritis.

2. Synovial fluid WBC framing

WHAT YOU SEE

Joint-fluid syringe/vial icons on the 'QUICK REFERENCE CARDS' stand — the aspirated tube of synovial fluid is the test whose WBC count tiers bland vs inflammatory vs septic.

WHAT IT ENCODES

Synovial fluid WBC count tiers the diagnosis: non-inflammatory <2,000/μL (OA, trauma; clear/yellow, viscous), inflammatory 2,000–50,000/μL (RA, gout, spondyloarthropathy; cloudy), and septic >50,000/μL (often >100,000 with >90% PMNs; purulent). Also always send Gram stain, culture, and crystal analysis under polarized light. Mnemonic: '2-50' — under 2k is bland, over 50k is bug.

BOARD TRAP

WBC >50,000 with PMN predominance = treat as septic until cultures return — answering 'gout flare' and giving only NSAIDs while a septic joint is destroyed in 24–48 h is the classic miss. Crystals and infection can coexist.

3. Acromegaly arthropathy

WHAT YOU SEE

Exhibit 1: large jaw/hands man, X-ray

WHAT IT ENCODES

Excess GH/IGF-1 in acromegaly drives cartilage and periarticular soft-tissue hypertrophy, so the EARLY X-ray shows WIDENED joint spaces — the opposite of usual OA — before secondary degenerative OA narrows them. Look for the gestalt: enlarged jaw/hands/feet, frontal bossing, carpal tunnel, and barrel chest. Screen with IGF-1, confirm with an oral glucose tolerance test (GH fails to suppress).

BOARD TRAP

Early WIDENED joint space (cartilage overgrowth), not the narrowing of primary OA, is the discriminator — choosing 'primary osteoarthritis' misses the treatable pituitary cause.

4. Hemochromatosis arthropathy

WHAT YOU SEE

Exhibit 2: '2nd/3rd MCP' hook osteophytes, ↑transferrin sat

WHAT IT ENCODES

Iron overload deposits in synovium producing arthropathy of the 2nd and 3rd MCP joints with characteristic HOOK-shaped osteophytes, plus secondary CPPD chondrocalcinosis. Screen with transferrin saturation (>45%) and ferritin; confirm HFE C282Y mutation. Look for the triad of diabetes, bronze skin, and cirrhosis. Treat with phlebotomy.

BOARD TRAP

MCP (2nd/3rd) involvement with iron studies points to hemochromatosis, NOT primary OA, which spares the MCPs and favors DIP/PIP — picking 'OA' overlooks an inherited, treatable systemic disease.

5. CPPD / chondrocalcinosis link

WHAT YOU SEE

A blue rhomboid crystal icon tucked under the hemochromatosis exhibit labeled 'CHONDROCALCINOSIS' — the calcium-pyrophosphate crystal linking iron overload to pseudogout.

WHAT IT ENCODES

CPPD (pseudogout) crystals are calcium pyrophosphate: rhomboid-shaped and POSITIVELY birefringent (blue when parallel to the slow axis). It causes acute knee/wrist flares and chondrocalcinosis on X-ray. New/young CPPD should trigger a search for the 4 H's: Hemochromatosis, Hyperparathyroidism, Hypomagnesemia, and Hypophosphatasia. Treat flares with NSAIDs, colchicine, or intra-articular steroids.

BOARD TRAP

CPPD = positively birefringent rhomboids; gout (monosodium urate) = NEGATIVELY birefringent needles (yellow when parallel) — swapping the two crystal descriptions is the single most tested polarized-microscopy trap.

6. Hemophilia arthropathy

WHAT YOU SEE

Exhibit 3: warm swollen knees, blood vial, Factor VIII

WHAT IT ENCODES

Hemophilia A (Factor VIII) or B (IX) deficiency causes recurrent hemarthrosis — most often knees, ankles, and elbows — and repeated bleeds drive synovial hypertrophy and a chronic destructive arthropathy. An acute bleed presents as a warm, tense, swollen joint. Treat acutely by replacing the deficient factor (target ~50% for joint bleeds), NOT routine aspiration.

BOARD TRAP

The warm swollen joint here is BLEEDING, not infection — the wrong move is to anticoagulate or aspirate-and-wait; give factor replacement first. (X-linked recessive, so think boys/men with prolonged aPTT and normal PT/platelets.)

7. Sickle cell — dactylitis

WHAT YOU SEE

Exhibit 4: child hands, 'DACTYLITIS 6mo-3yr'

WHAT IT ENCODES

Vaso-occlusion of the small bones of the hands and feet causes dactylitis (hand-foot syndrome), the classic FIRST pain crisis in infants/toddlers age 6 months–3 years. Repeated infarction also causes avascular necrosis of the femoral head. Management is hydration, analgesia, and hydroxyurea for prevention.

BOARD TRAP

Sickle-cell joint pain is usually vaso-occlusive or AVN, but a single hot joint still demands aspiration to exclude septic arthritis/osteomyelitis — assuming 'just a pain crisis' and missing infection is the trap.

8. Salmonella osteomyelitis

WHAT YOU SEE

A small 'SALMONELLA' / osteomyelitis tag on the sickle-cell pedestal (exhibit 4) — flags the bug that classically seeds infarcted sickle bone.

WHAT IT ENCODES

Functional asplenia in sickle cell predisposes to osteomyelitis, and the classically tested organism is Salmonella, which seeds infarcted bone. Diagnose with MRI and bone/blood cultures. Note the boards nuance: across all comers (including sickle cell), Staphylococcus aureus remains the single most common cause overall.

BOARD TRAP

Salmonella is the CLASSIC sickle-cell association, but if the question asks 'most common cause of osteomyelitis overall' the answer is S. aureus — picking the wrong one depends on reading exactly what is asked.

9. Charcot joint (neuropathic)

WHAT YOU SEE

Exhibit 5: man with deformed foot, rocker bottom

WHAT IT ENCODES

Charcot (neuropathic) arthropathy results from loss of protective sensation: unperceived repetitive microtrauma destroys the joint, producing a grossly deformed, often 'rocker-bottom' foot with the destruction centered on the tarsal/tarsometatarsal joints. The #1 cause is diabetic neuropathy (others: syringomyelia, tabes dorsalis, leprosy). Treat with offloading/total-contact casting.

BOARD TRAP

The hallmark discriminator is destruction GROSSLY out of proportion to the minimal or absent pain — it is neuropathic, not inflammatory or infectious; a swollen warm diabetic foot may be Charcot rather than (or alongside) osteomyelitis.

10. Hypertrophic osteoarthropathy / clubbing

WHAT YOU SEE

Exhibit 6: clubbed fingers, periostitis X-ray

WHAT IT ENCODES

Hypertrophic osteoarthropathy is the triad of digital CLUBBING, periostitis of long bones (X-ray shows periosteal new bone), and symmetric arthralgia. Secondary HOA is strongly associated with intrathoracic malignancy (especially non-small-cell lung cancer) and other chronic lung/cardiac disease. The Schamroth sign (loss of the nailbed window) screens for clubbing.

BOARD TRAP

New clubbing plus bone pain mandates chest imaging to hunt for lung malignancy — attributing it to 'aging' or 'OA' and skipping the CXR/CT misses the underlying cancer.

11. PVNS

WHAT YOU SEE

Exhibit 7: PVNS, brownish synovium, MRI

WHAT IT ENCODES

Pigmented villonodular synovitis is a benign but locally aggressive proliferation of synovium, usually MONOARTICULAR in the knee, producing hemosiderin-laden brown synovium and a recurrent BLOODY (serosanguineous) effusion. MRI shows characteristic blooming/low-signal from hemosiderin and is diagnostic; treatment is surgical synovectomy.

BOARD TRAP

Chronic atraumatic BLOODY monoarthritis of the knee is PVNS, not 'recurrent injury' or simple hemarthrosis — anchoring on trauma and missing the MRI hemosiderin signature is the error.

12. Symmetric vs asymmetric pattern

WHAT YOU SEE

The rows of hand exhibits contrasting both-sides-alike joints versus one-sided joints — the drawn hands map symmetric small-joint (RA/SLE) against asymmetric (spondyloarthropathy) patterns.

WHAT IT ENCODES

Symmetric small-joint polyarthritis (MCPs/PIPs/wrists) points to rheumatoid arthritis or SLE; asymmetric large-joint or oligoarticular disease points to the spondyloarthropathies (psoriatic, reactive, ankylosing) or crystal disease. RA spares the DIPs whereas OA and psoriatic arthritis classically involve them. Mnemonic: symmetric + small = RA.

BOARD TRAP

DIP involvement argues AGAINST RA and FOR OA or psoriatic arthritis — labeling DIP nodules (Heberden's) as 'rheumatoid' confuses degenerative with inflammatory disease.

13. Acute vs chronic timing

WHAT YOU SEE

The left-to-right progression of exhibits and their X-rays from early to destroyed joints — the timeline drawn across the museum encodes acute (<6 wk) versus chronic (>6 wk) disease.

WHAT IT ENCODES

Acute arthritis lasting <6 weeks is most often crystal, septic, or reactive/viral; chronic arthritis >6 weeks suggests RA, OA, or a spondyloarthropathy. Pairing acuity with joint count and inflammation narrows the differential quickly (e.g., acute + mono + inflammatory = septic/crystal). The 6-week mark is also the RA-classification threshold.

BOARD TRAP

Acute hot MONOarthritis is septic or crystal until arthrocentesis proves otherwise — calling it an OA flare and withholding aspiration is the recurring tested mistake.

14. Mono vs poly framing

WHAT YOU SEE

The museum row lining up single-joint exhibits against multi-joint exhibits — one pedestal-one-joint versus many-joints visually forks the differential.

WHAT IT ENCODES

Monoarthritis differential: septic arthritis, crystal disease (gout/CPPD), trauma/hemarthrosis, PVNS, and Charcot. Polyarthritis differential: RA, viral (parvovirus B19, hepatitis), SLE, and psoriatic arthritis. The number of joints involved is the first fork that, combined with inflammation status, drives the workup. Mnemonic: one hot joint = aspirate.

BOARD TRAP

A single acutely inflamed joint must be aspirated to rule out infection BEFORE assuming a known disease flare — septic arthritis can destroy cartilage within 1–2 days, so empiric NSAIDs without a tap is the wrong answer.
